

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**23CSE304 Embedded Systems Laboratory**

**Activity Sheet #2**

**Activity Name:** Switch Interfacing

|                          |                         |
|--------------------------|-------------------------|
| <b>Name of Student:</b>  | <b>Agneay B Nair</b>    |
| <b>Registration No.:</b> | <b>CH.SC.U4CSE24102</b> |
| <b>Activity Date:</b>    | <b>18-08-2026</b>       |
| <b>Submission Date:</b>  | <b>18-08-2026</b>       |

**Assessment Rubrics:**

| <b>Component</b>  | <b>Maximum Marks</b> | <b>Marks Allotted</b> |
|-------------------|----------------------|-----------------------|
| Program           | 3                    |                       |
| Output            | 4                    |                       |
| Viva              | 2                    |                       |
| Timely Submission | 1                    |                       |
| <b>Total</b>      | <b>10</b>            |                       |

**Faculty In-charge**

**Write an embedded C program for ARM architecture to interface switches. Debug and verify the program using IDE on a development computer system.**

## **CODE:**

```
#include<LPC213x.h>

void delay(int n){
    int i;
    for(i = 0;i<n*1000;i++);
}

int main(){
    PINSEL0 = 0x00000000;
    IODIR0 |= (1<<16);
    IODIR0 &= ~(1<<8);

    while(1){
        if(IOPIN0 & (1<<8)){
            // turn LED on
            IOSET0 = (1<<16);
            delay(10);
        }else{
            IOCLR0 = (1<<16);
            delay(10);
        }
    }
}
```



First.c

```
1  #include<LPC213x.h>
2
3  void delay(int n){
4      int i;
5      for(i = 0;i<n*1000;i++);
6  }
7
8  int main(){
9      PINSEL0 = 0x00000000;
10     IODIR0 |= (1<<16);
11     IODIR0 &= ~(1<<8);
12
13     while(1){
14         if(IOPIN0 & (1<<8)){
15             // turn LED on
16             IOSET0 = (1<<16);
17             delay(10);
18         }else{
19             IOCLR0 = (1<<16);
20             delay(10);
21         }
22     }
23 }
```

**OUTPUT:**

C:\Projects\ARM7-LPC2148-Assembly-Programs\Activity Sheet - 1\Code\First Program.uvproj - µVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

| Register       | Value      |
|----------------|------------|
| R0             | 0x00000000 |
| R1             | 0x00000000 |
| R2             | 0x00000000 |
| R3             | 0x00000000 |
| R4             | 0x00000000 |
| R5             | 0x00000000 |
| R6             | 0x00000000 |
| R7             | 0x00000000 |
| R8             | 0x00000000 |
| R9             | 0x00000000 |
| R10            | 0x00000000 |
| R11            | 0x00000000 |
| R12 (SP)       | 0x00000000 |
| R14 (LR)       | 0x00000000 |
| R15 (PC)       | 0x00000000 |
| CPSR           | 0x00000000 |
| SPSR           | 0x00000000 |
| User/System    |            |
| Fast Interrupt |            |
| Interrupt      |            |
| Supervisor     |            |
| Abort          |            |

Disassembly

```
9: PINSEL0 = 0x00000000;
0x000001BC E3A00000 MOV R0,#0x00000000
0x000001C0 E59F106C LDR R1,[PC,#0x006C]
0x000001C4 F5810000 STR R0,[R1]
```

FirstC

```
4 int i;
5 for(i = 0; i < n*1000; i++);
6 }
7
8 int main() {
9 PINSEL0 = 0x00000000;
10 IODIR0 |= (1<<16);
11 IODIR0 &= ~(1<<8);
12
13 while(1) {
14 if(IOPIN0 & (1<<8)) {
15 // turn LED on
16 IOSET0 = (1<<16);
17 delay(10);
18 } else {
19 IOCLR0 = (1<<16);
20 }
```

Command

Running with Code Size Limit: 32K  
Load "C:\Projects\ARM7-LPC2148-Assembly-Programs\Activity Sheet - 1\Code\

Call Stack + Locals

| Name | Locati... | Type   |
|------|-----------|--------|
| n    | 0x0000... | int f0 |

ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess

Real-Time Agent: Target Reset Simulation t1: 0.00007217 sec L:9 C:1 CAP: NUM SCRL OVR: R/W

